



DEPI IBM Data Science

Employee Attrition Prediction & Analysis Project

Course instructor:

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1. Project Proposal

1.1 Project Overview:

The Employee Attrition Prediction and Analysis project focuses on building a machine learning model to predict employee turnover (attrition) within an organization. By identifying employees who are likely to leave, companies can take proactive measures to improve retention. The project follows a data science lifecycle, from data collection and exploration to model deployment and monitoring, aimed at improving organizational retention strategies.

1.2 Objectives:

1. **Data Preparation:** Collect, explore, and preprocess employee datasets (from open repositories like Kaggle or UCI) to establish a clean foundation for analysis.
2. **Advanced Analysis & Feature Engineering:** Perform in-depth statistical analysis and create advanced features (e.g., tenure categories, salary bands) to uncover key drivers of attrition.
3. **Model Development:** Build, train, and optimize robust classification models (such as Random Forest or XGBoost), ensuring proper handling of class imbalance using techniques like SMOTE.
4. **Deployment & MLOps:** Implement MLOps practices (e.g., MLflow) for tracking, and deploy the final model as an API (Flask/Fast API) coupled with an interactive HR dashboard (Streamlit/Dash) for real-time predictions.
5. **Business Value Presentation:** Deliver comprehensive documentation and an engaging presentation for stakeholders, translating model predictions into actionable HR retention strategies.

1.3 Project Scope:

1. Utilizing open-source employee datasets to train the machine learning models
2. Developing binary classification models to predict attrition vs. non-attrition.
3. Designing an interactive dashboard (e.g., Streamlit or Dash) for HR teams to input data and receive real-time attrition risk scores
4. Deploying the model via a REST API and implementing basic MLOps monitoring.

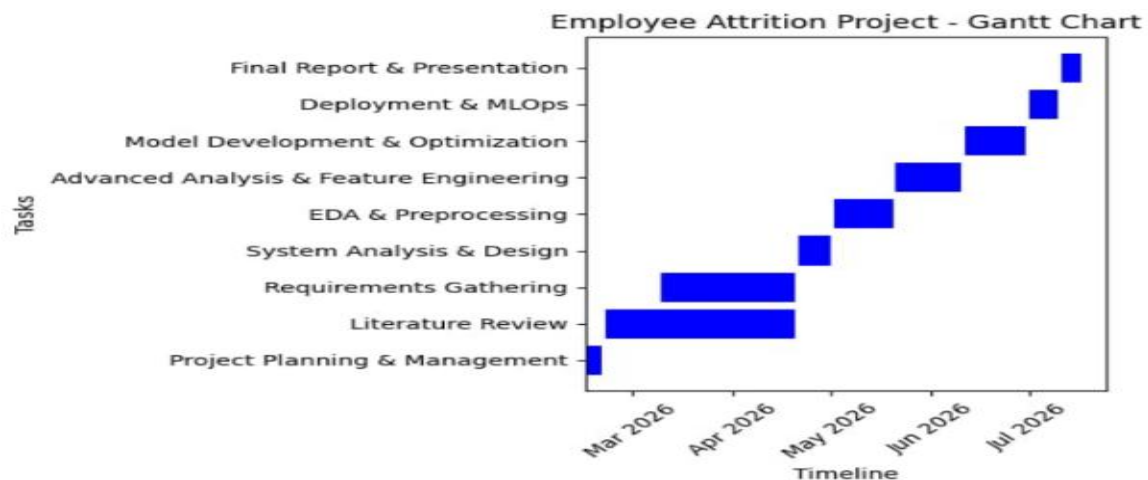
2. Project Planning

This section outlines the project timeline, milestones, deliverables, and resource allocation plan for the Employee Attrition Prediction project.

1.1 Project Timeline & Milestones

Phase	Start Date	End Date	Key Deliverable
Project Planning & Management	15/2/2026	20/2/2026	Proposal & Project Plan
Literature Review	21/2/2026	20/4/2026	Literature Review Report
Requirements Gathering	10/3/2026	20/4/2026	Requirements Document
System Analysis & Design	21/4/2026	1/5/2026	System Design Document
Implementation (EDA + Modeling)	2/5/2026	10/7/2026	Trained & Tuned Model
Deployment & MLOps	1/7/2026	10/7/2026	Deployed API & Monitoring
Final Presentation & Reports	11/7/2026	17/7/2026	Final Report & Presentation

Gantt Chart



1.2 Resource Allocation

Role	Responsibility
Project Manager	Timeline management, coordination, risk tracking
Data Engineer	Data cleaning, preprocessing, data validation
Data Analyst	EDA, visualization, reporting insights
ML Engineer	Model training, tuning, evaluation
MLOps Engineer	Deployment, monitoring, version control
Documentation Lead	Reports, presentation, GitHub documentation

3.Risk Assessment & Mitigation Plan

This section identifies potential risks that may impact the project's execution, evaluate their probability and impact, and define mitigation strategies. Risks will be reviewed bi-weekly during sprint meetings.

Top 5 Risks:

Risk ID	Risk Description	Impact	Probability	Mitigation Plan	Owner
R1	Imbalanced dataset	High	High	Use SMOTE and monitor recall	ML Engineer
R2	Model overfitting	High	Medium	Apply cross-validation and regularization	ML Engineer
R3	Poor data quality	Medium	High	Data cleaning and validation checks	Data Engineer

R4	Deployment failure	High	Medium	Test locally before production deployment	MLOps Engineer
R5	Timeline delay	High	Medium	Weekly monitoring and buffer time	Project Manager

Other Risks:

1- Data & Technical Risks

Even the best algorithm is only as good as the data you feed it.

The "Garbage In, Garbage Out" Factor: If your historical data is messy or missing key details (like why someone left), your predictions will be guessing at best.

Data Leakage: This is a common "oops" in data science. If you include data points that only happen after someone decides to leave (like "unused vacation days" or "exit interview score"), the model will look perfect in testing but fail in real life.

The "Vibe" Problem: Quantitative data (salary, tenure) is easy to track, but qualitative data (a bad boss, burnout, or personal issues) is hard to capture. Your model might miss the human element.

2. Ethical & Privacy Risks

Since you are dealing with people's livelihoods, the stakes are high.

Selection Bias: If the company historically promoted one group over another, the AI might learn that the "neglected" group is more likely to leave. Following the model blindly could reinforce unfair patterns.

Privacy Concerns: Employees might feel "watched." If they know an AI is judging their likelihood of quitting based on their behavior, it can damage trust and actually increase attrition.

The "Self-Fulfilling Prophecy": If a manager sees a high attrition risk score for an employee, they might stop investing in that person's growth, which eventually causes them to quit.

3. Organizational & Action Risks

A model is useless if the company doesn't know how to use the results.

Misinterpretation: A "70% risk" doesn't mean the person is definitely quitting tomorrow. Managers might panic and take the wrong actions.

Lack of Actionable Insight: Knowing who is leaving is great, but if the model doesn't tell you why, you can't fix the problem. Offering a raise to someone who is leaving because of a toxic culture won't work.

Model Decay: People's reasons for quitting change (e.g., the shift to remote work in 2020). If you don't update the model, it will become obsolete quickly.

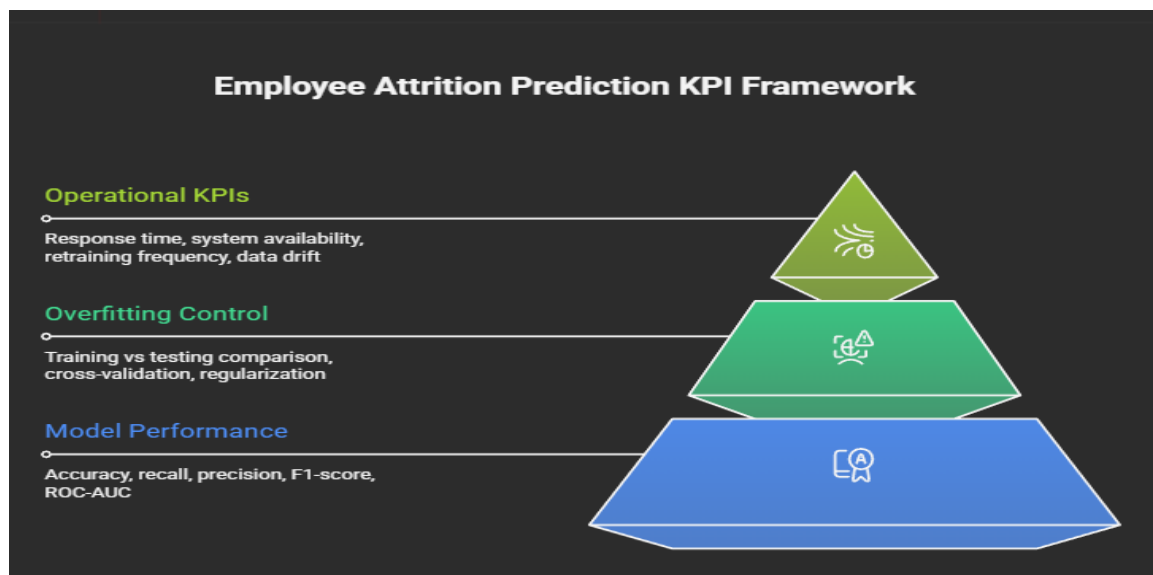
4. Key Performance Indicators (KPIs)

This section defines the Key Performance Indicators (KPIs) used to evaluate the performance and effectiveness of the Employee Attrition Prediction model.

The KPIs focus on measuring model accuracy, generalization capability, and overall predictive strength.

A KPI framework diagram is provided to illustrate the relationship between model performance, operational reliability, and overall system effectiveness.

4.1 KPI Framework Overview



The KPI framework categorizes evaluation metrics into:

- Model Performance KPIs
- Business-Oriented KPIs
- Operational KPIs

This structure ensures that the system is evaluated from both a technical and practical perspective.

4.2 Model Performance KPIs

Since this is a Machine Learning project, model performance metrics represent the primary evaluation criteria.

Accuracy

Measures the overall correctness of the model's predictions across both classes.

Recall

Evaluates the model's ability to correctly identify employees who are likely to leave. This metric is particularly important in attrition prediction to minimize false negatives.

Precision

Measures the proportion of correctly predicted attrition cases among all predicted attrition cases.

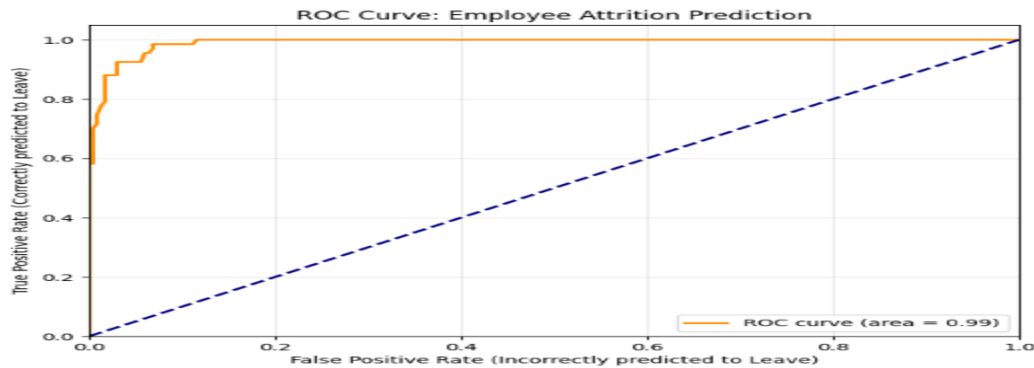
Score

Provides a balanced measure between Precision and Recall.

ROC-AUC

Represents the model's ability to distinguish between employees who will leave and those who will stay.

Higher AUC values indicate better classification performance.



The ROC curve evaluates the model's ability to distinguish between employees who are likely to leave (Attrition = 1) and those who will stay (Attrition = 0).

The model achieved an AUC score of **0.99**, which indicates excellent classification performance and very strong discriminative capability.

This means the model can effectively separate high-risk employees from low-risk employees across different classification thresholds.

However, further validation is necessary to ensure that this high performance is not due to overfitting.

4.3 Overfitting Control

To ensure proper generalization:

- The difference between training and testing performance is monitored.
- Cross-validation techniques are applied.
- Regularization methods are used where necessary.

This ensures that the model performs consistently on unseen data.

4.4 Operational KPIs

Although the project focuses on machine learning performance, operational reliability is also considered:

- Prediction response time
- System availability
- Model retraining frequency
- Data drift monitoring

These ensure long-term stability of the prediction system.